

Bayesian rows for the modified-Cholesky EnKF: climatological priors and adaptive memory in the precision estimate

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Abstract

The modified-Cholesky ensemble Kalman filter (EnKF-MC) estimates the background precision $\mathbf{B}^{-1} = \mathbf{L}^\top \mathbf{D} \mathbf{L}$ one row at a time, each row being a regression of a state component on its predecessors within a localization radius, fitted to the N ensemble anomalies with a ridge penalty. We treat each row as a Bayesian regression and change two things about its prior. First, the prior mean is not zero: at the start of a cycled experiment it is the coefficient vector of the same regression fitted to a climatology of the model, and from the second cycle on it is the previous cycle's posterior. Second, the prior weights are not fixed: the forgetting factor of the sequential posterior and the weight of the climatological term are chosen for every row and every cycle by maximising the marginal likelihood of the new ensemble anomalies. The result is a precision whose coefficients are estimated from an effective sample far larger than N where the forecast error is stationary, and from the current ensemble alone where it changes. On the quasi-geostrophic testbed of Sakov and Oke with $N = 40$ members, the Bayesian rows reduce the analysis error of the plain EnKF-MC at every cycle of a 36-cycle run (by 5–20%), reduce the error of the first analysis from a climatological ensemble by 6–9% at every observation density between 1% and 40%, and end four times below the best LETKF. The hyperparameters selected by the evidence have a transparent physical reading: the climatology is used over the jet and released after one analysis; memory is kept where the flow is quiescent and dropped along the jet. We also report, with the same care, what does not work: five ways of choosing *which* predecessors a row uses, from the ensemble, from the dynamics or from the climatology, none of which beats a uniform radius once the filter has converged; the gain is in the values of the rows and in how much data they see, not in their sparsity pattern.

1 Introduction

Localization in ensemble Kalman filters exists because N members cannot estimate an $n \times n$ covariance. The modified-Cholesky EnKF (EnKF-MC) of Nino-Ruiz and co-workers localizes in the precision: component i is regressed on its predecessors $P(i)$, the components with a lower index within a radius r , and the regression coefficients fill row i of a unit lower-triangular $\mathbf{L}$. With a ridge penalty, the row is a well-posed problem even when $|P(i)| > N$, the precision is sparse, and the analysis is a sparse linear solve. The radius is the localization parameter; it is chosen by a sweep, and Bickel and Levina's theory of banded Cholesky estimators says why a small radius is right when N is small: the bandwidth balances the bias of truncating the true dependence against the variance of estimating it from N samples.

This paper does not change the sparsity pattern. It changes what the rows are fitted to. Two observations motivate it. The first is that the ridge penalty of a row is a Gaussian prior on its coefficients, centred at zero; nothing forces the centre to be zero, and at the first analysis of a cycled experiment, when the ensemble is a sample of the model's climatology, the coefficients of the same

regression fitted to that climatology are the correct prior mean. The second is that the forecast error at one cycle resembles the forecast error at the previous one, so the previous row is a useful prior for the current one; how useful depends on how fast the error is changing, which differs from place to place and can be read from the data. Both observations lead to the same construction: a sequential Bayesian regression per row, whose hyperparameters – a forgetting factor and a climatological weight – are chosen by the marginal likelihood of the new anomalies.

Section ?? gives the method. Section ?? the testbed. Section ?? the results: first analysis, cycled runs, comparison with the LETKF, and the maps of what the evidence decides. Section ?? reports the structural alternatives that were tried and why they lose. Section ?? discusses limits and what is left to establish.

2 Method

2.1 The modified-Cholesky row as a regression

Let $\Delta\mathbf{X} \in \mathbb{R}^{n \times N}$ be the forecast anomalies. For component i with predecessors $P(i)$ ($p = |P(i)|$), write $\mathbf{y} \in \mathbb{R}^N$ for its anomalies and $\mathbf{X} \in \mathbb{R}^{N \times p}$ for those of its predecessors. The row model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}), \quad (1)$$

and $\mathbf{L}_{ii} = 1$, $\mathbf{L}_{i,P(i)} = -\boldsymbol{\beta}^\top$, $\mathbf{D}_{ii} = 1/\sigma^2$, $\hat{\mathbf{B}}^{-1} = \mathbf{L}^\top \mathbf{D} \mathbf{L}$. The standard estimator is ridge regression,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X} + a\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}, \quad a = \alpha \operatorname{tr}(\mathbf{X}^\top \mathbf{X})/p, \quad (2)$$

with α dimensionless and the same for every row. This is the posterior mean of $\boldsymbol{\beta}$ under the prior $\boldsymbol{\beta} \sim \mathcal{N}(0, \sigma^2/a \mathbf{I})$.

2.2 Bayesian rows

We keep the likelihood and change the prior. All statistics are kept in correlation scale – $\mathbf{X}$ and $\mathbf{y}$ standardised by their ensemble standard deviations – so that cycles with different spread contribute equally; coefficients are rescaled at the end.

Sequential prior. Let $\mathbf{G}_t = \mathbf{X}_t^\top \mathbf{X}_t/N$ and $\mathbf{b}_t = \mathbf{X}_t^\top \mathbf{y}_t/N$ be the sufficient statistics of cycle t . The row keeps a memory

$$\mathbf{G}_t^{\text{mem}} = \rho_t \mathbf{G}_{t-1}^{\text{mem}} + \mathbf{G}_t, \quad \mathbf{b}_t^{\text{mem}} = \rho_t \mathbf{b}_{t-1}^{\text{mem}} + \mathbf{b}_t, \quad (3)$$

with forgetting factor $\rho_t \in [0, 1)$: the posterior of one cycle is the prior of the next, discounted.

Climatological term. From M independent states of the model’s climatology the same statistics are computed once, $\mathbf{G}^c$, $\mathbf{b}^c$, weighted by $w_0 = M/N$: the climatology is worth as many cycles as ensembles fit in it. At cycle t the row’s prior is $\mathbf{G}_t^{\text{mem}} + w_t \mathbf{G}^c$, $\mathbf{b}_t^{\text{mem}} + w_t \mathbf{b}^c$ with $w_t \in [0, w_0]$.

Estimator. The row coefficients are

$$\hat{\boldsymbol{\beta}}_t = (\mathbf{G}_t^{\text{mem}} + w_t \mathbf{G}^c + a \mathbf{T})^{-1} (\mathbf{b}_t^{\text{mem}} + w_t \mathbf{b}^c), \quad \mathbf{T} = \operatorname{diag}((1 + \tau d_{ij})^2), \quad (4)$$

where d_{ij} is the Chebyshev distance from i to predecessor j and τ a tapering slope: the penalty of a coefficient grows with the distance of its predecessor, a soft version of the banded Cholesky. The residual variance σ^2 is that of the current ensemble under $\hat{\boldsymbol{\beta}}_t$, so the scale of $\hat{\mathbf{B}}^{-1}$ follows the ensemble while its shape draws on the memory and the climatology.

Hyperparameters by evidence. ρ_t and w_t are chosen, per row and per cycle, to maximise the marginal likelihood of the new anomalies under the prior:

$$(\rho_t, w_t) = \arg \max \log \mathcal{N}(\mathbf{y}_t; \mathbf{X}_t \boldsymbol{\beta}^{\text{prior}}, \sigma^2 (\mathbf{I} + \mathbf{X}_t \mathbf{P}^{-1} \mathbf{X}_t^\top)), \quad \mathbf{P} = N(\rho \mathbf{G}_{t-1}^{\text{mem}} + w \mathbf{G}^c), \quad (5)$$

over a small grid ($\rho \in \{0.2, 0.4, 0.6, 0.8, 0.95\}$, $w/w_0 \in \{0, 0.1, 0.3, 1\}$), with σ^2 profiled out. Each evaluation is one $N \times N$ Cholesky; the cost per cycle is a small multiple of the plain EnKF-MC. The evidence penalises prior complexity (log det term) and does not reward fitting noise, unlike cross-validation; it uses only forecast anomalies, never observations or truth.

What the evidence cannot choose. The tapering slope τ (and the base ridge α) are localization hyperparameters: they protect the *analysis* from noisy long-range coefficients, and the row likelihood does not see the analysis. Chosen by the row evidence they go to zero and the filter degrades (measured, Section ??). They are therefore treated like the radius and the inflation: fixed by a sensitivity sweep.

Structure. The predecessor set is the square of radius 2 (the optimum of the uniform sweep at $N = 40$) united with the climatological predecessors selected by a lasso row regression on the M climate states (window 6, universal-threshold penalty with $c = 0.5$). With the tapered ridge the long links cost nothing once the climatology has been released; without it they do (Section ??).

3 Testbed

Quasi-geostrophic model of Sakov and Oke as implemented in `pyteda`, 49×49 grid, dissipation ten times the default so that a stationary regime exists, state q only, Dirichlet boundary excluded from estimation ($47 \times 47 = 2209$ components), $dt = 0.3125$, 20 time units between analyses. Climatology: spin-up to $t = 20000$ then 200 snapshots every 250 units (81 every 200 in the development runs); ensemble members and truth are independent draws from it. Observations of q (or of ψ , Sakov and Oke’s choice, through the Helmholtz inverse) with 5% of the climatological spread as error, on a fixed lattice of stride 2 (24% of the interior) or on random networks observing 1–40% of it. Multiplicative inflation 1.05, calibrated for the plain EnKF-MC; $\alpha = 0.3$; $N = 40$. Errors are RMSE of q over the interior in units of the climatological standard deviation, so that 1 is “as bad as not assimilating”.

4 Results

4.1 First analysis from a climatological ensemble

At the first analysis the background error *is* the climatology, and the climatological prior mean is exact. Table ?? gives the ratio of analysis to background RMSE for five seeds, observing ψ , random network at 25%.

The dense climatological covariance of the same 81 states, used as an oracle $\mathbf{B}$ in a Kalman analysis, gives 0.894 at 1% density where the method gives 0.937 and the best uniform row 0.971: the information limit of 22 observations is close, and the method recovers about half of what separates the plain row from it.

4.2 Cycled runs

Figure ?? shows 34 cycles on the lattice. The plain EnKF-MC with $r = 2$ converges to 0.11; the LETKF, at its best radius, to 0.40; the Bayesian rows to 0.10, below the plain rows at every cycle

structure	ridge→ 0	ridge→ β^c , $\alpha = 100$
uniform $r = 1$	0.707	0.691
uniform $r = 2$	0.738	0.684
uniform $r = 3$	0.760	0.669
uniform $r = 4$	0.817	0.658
climate lasso $w6$	0.709	0.660
LETKF $r = 1$	0.792	–

Table 1: First analysis, ψ observed, random network 25%, $N = 40$, five seeds. With the climatological prior the ordering of the radii inverts: the wide row, useless when estimated from 40 members, is the best when its coefficients come from 81 climate states. The gain over the best plain row is 7%. Across densities 1–40% and both observed variables the climatological prior is first at every density (2–9%).

(ratio 0.80–0.95). Without the tapered ridge, the climatological long links help at cycle 0 and then cost a plateau (0.71→0.72→0.72) before recovering to 0.16; with it the descent is monotone from the best start.

../results/figures_{dev}/bayes_{rows}_{taper}.png

Figure 1: Analysis RMSE per cycle, QG, $N = 40$, lattice 24%, inflation 1.05, one seed.

4.3 What the evidence decides

Figure ?? maps the hyperparameters chosen row by row. At cycle 0 the climatological weight is at its maximum over the jet and the eddies and already released in the quiescent north; from cycle 1 it is near zero everywhere but a few rows. The forgetting factor is high (memory kept) in the north and low (forget) along the jet. The implied background correlation of a point passes from the climatological bands at cycle 0 to a local blob at cycle 15 with the same twelve neighbours in the row: the values and the memory do the work, not the pattern.

../results/figures_{dev}/bayes_{rows}_{structure}.png

Figure 2: Flow, climatological weight chosen at cycles 0 and 15, forgetting factor at cycle 15, and the implied correlation of one point at cycles 0 and 15.

4.4 Sensitivity and what is fixed by hand

ρ and w are chosen by the evidence; $w_0 = M/N$ is a definition. Fixed: $\alpha = 0.3$, $r = 2$, inflation 1.05 (inherited, sensitivity in the supplement), $\tau = 0.3$ ($\tau = 1$ over-regularises: 0.20 at cycle 33; $\tau = 0$ leaves the long links noisy), and the climate lasso window and penalty (insensitive with the taper). Choosing τ by the row evidence selects $\tau = 0$ and the run ends at 0.25: the evidence of a row is not the right criterion for a localization parameter.

5 What does not work, and why

Five ways of choosing *which* predecessors a row uses were tried on the same testbed, each cycled for at least 36 cycles: a variational cost per component (which selects the *least* correlated candidate, a defect

of its algebra), the partial correlations of a wide probe precision, the trajectories of the ensemble-mean flow, the lagged cross-correlation between analysis and forecast ensembles (an estimate of the propagator), and a lasso row selection on the ensemble. All of them beat the uniform square on a single analysis and none beats it after ten cycles. The reason is visible in a perturbation experiment: a point perturbation of q does not travel along the flow within an assimilation interval, or within ten of them; it grows in place. The converged forecast error is local, fast-decaying, and $r = 2$ is the Bickel–Levina optimum for it at $N = 40$. Structures richer than that pay selection noise without removing bias. The climatological structure is the one exception, and only because the climatology, not the ensemble, supplies its coefficients.

6 Discussion

The method is a hybrid of climatological and ensemble information built inside the precision, row by row, without a dense climatological covariance, without a blending weight for covariances, and with the blending decided by the data where it can be. Its cost is a small multiple of the EnKF-MC. Its gains are in the first analyses, in the transient, and modestly in the converged state; they are measured on one seed and one network and must be confirmed on sparse random networks with the LETKF alongside, which is the experiment in progress. The evidence chooses memory and climatological weight well; it cannot choose the tapering, and we do not claim it can.